

# ICRA 2025 Competition Proposal: The Earth Rover Challenge 3

## Description

With the rise of generalist foundation models for robot navigation [1,2], new frontiers involving challenging open-world and open-vocabulary mobility scenarios are now of immense interest to the robotics community.

Our competition aims to explore the possibility of *globally distributed in-the-wild* navigation evaluations at an unprecedented scale, along with the release of a substantial real-world navigation dataset collected from 30+ cities (>7k hours).

Leveraging on (a) a large fleet of outdoor navigation robots deployed across multiple cities [3], (b) an indoor navigation environment at NYU & (c) an off-road environment at GMU, we aim to study whether state of the art open-world autonomous navigation models are able to effectively operate across all 3 settings and how they fare against human teleoperated performance under the same environments.

We propose a distributed competition across 3 complementary domains of navigation:

- Indoor Navigation (New York University): missions in structured indoor buildings, using image goals
- Off-Road Terrain Navigation (George Mason University): missions on natural, uneven terrain with image goals
- In-the-Wild Urban Navigation (Multiple Cities worldwide): outdoor city navigation with GPS goals

[1] <https://general-navigation-models.github.io/>

[2] <https://ovmm.github.io/>

[3] <https://www.frodobots.ai/>

## Competition Rules

### Navigation Missions

The goal of the competition is to remotely control small sidewalk robots in order to complete various navigation missions in 3 settings: indoor, off-road, and outdoor urban environments in real time.

Each mission consists of a series of checkpoints that the robot needs to navigate through. Checkpoints are defined differently depending on the environment:

- Indoor: image goals
- Off-road: image goals
- Outdoor: GPS coordinates

Every mission is given certain award points: 1-10 based on the difficulty, e.g. whether crossing roads, crowded space (see table 1), and is considered “mission complete” when the robot returns to the end point of the given mission after registering at various checkpoints.

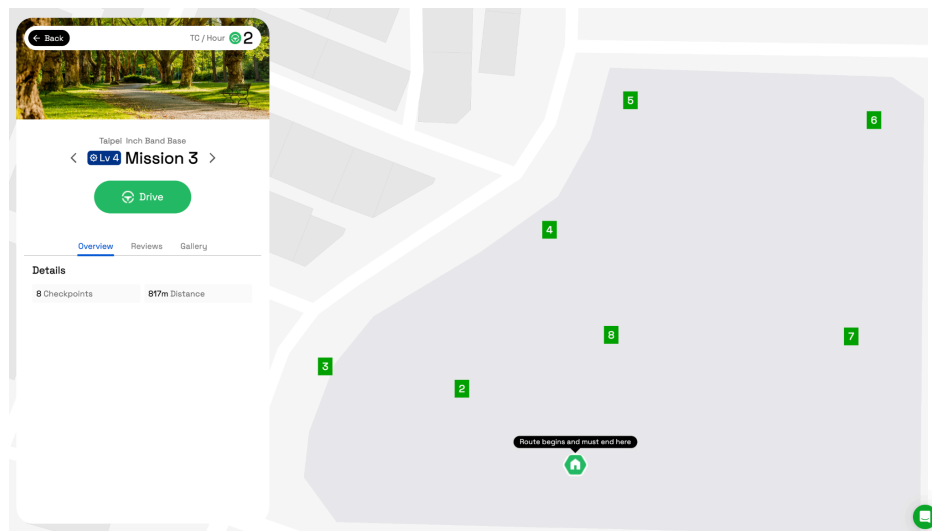


Figure 1: Example of a navigation mission (in Taipei)

| Points | Crossing Roads                                           | Pedestrian Traffic         | Distance (Checkpoints) | Terrain                         | Navigation           | Path                               |
|--------|----------------------------------------------------------|----------------------------|------------------------|---------------------------------|----------------------|------------------------------------|
| 1      | No                                                       | No people                  | 2                      | Smooth wide path                | Straight path        | Enclosed park                      |
| 2      | No                                                       | Few people                 | 3-4                    | Smooth narrow path              | Straight path        | Rural area                         |
| 3      | No                                                       | Some people                | 4-5                    | Smooth wide path (with curbs)   | 1 turn               | Beside a hill                      |
| 4      | No                                                       | Groups of People           | 5-7                    | Smooth narrow path (with curbs) | 2-3 turns            | Beside water body                  |
| 5      | No                                                       | Kids                       | 7-9                    | Requires going up/down hill     | 3-4 turns            | Beside parking lot                 |
| 6      | Road crossing in rural areas with no cars                | Cyclists, scooters, bikers | 10-12                  | Muddy or snowy                  | 4-5 turns            | Bicycle path                       |
| 7      | Road crossing with pedestrian crossing with few cars     | Cyclists, scooters, bikers | 12-24                  | Rocky paths with potholes       | >6 turns             | Path by small road with bikes/cars |
| 8      | Road crossing with traffic light and pedestrian crossing | Few nearby cars            | >14                    | Tall grass fields               | Alleys, small path   | Path by highway                    |
| 9      | Crossing 2 roads & above                                 | Nearby cars                | >14 with crossings     | Tall grass fields with potholes | Off route navigation | On road with cars                  |

|    |                                                            |                  |                             |                        |                        |                                          |
|----|------------------------------------------------------------|------------------|-----------------------------|------------------------|------------------------|------------------------------------------|
| 10 | Crossing road with no pedestrian crossing or traffic light | Many nearby cars | >18 with multiple crossings | Road with cars passing | Traffic light spotting | On road with cars in different direction |
|----|------------------------------------------------------------|------------------|-----------------------------|------------------------|------------------------|------------------------------------------|

Table 1: Difficulty scoring metrics of a navigation mission

The team whose navigation model achieves the highest aggregate award points will win the competition. If multiple teams have the same aggregate award points, the amount of time to complete missions is used to break the tie.

### Robot Platform

Each Earth Rover unit weighs less than 2 kg and moves at a max speed of ~3 Km/Hr. It is able to turn on the spot and comes equipped with front/back cameras, 4G connection, GPS/RTK & IMU. It has limited edge computing and is meant to be 100% remotely controlled by human drivers or AI navigation models hosted on the cloud.



Figure 2: Photos of an Earth Rover unit



Figure 3: Example screenshots of driving footage of Earth Rovers currently deployed in 30+ cities

## Competition Schedule

We plan to have multiple missions spanning global outdoor sites, an indoor track at NYU and an off-road track at GMU.

The competition consisting of all missions will be conducted in sequence over the course of 2 days, and live-streamed back to the conference center for the audience to watch. Here is a planned schedule for both days:

| Time          | Day 1                                                          | Day 2                                             |
|---------------|----------------------------------------------------------------|---------------------------------------------------|
| 08:30 - 09:00 | Participants check in                                          |                                                   |
| 09:00 - 10:00 | Participants prepare and test their connectivity, policy, etc. |                                                   |
| 10:00 - 11:00 | Official competition: 1 Round of Remote Mission                |                                                   |
| 11:00 - 12:00 | Official competition: 1 Round of Remote Mission                |                                                   |
| 12:00 - 13:30 | Lunch                                                          |                                                   |
| 13:30 - 14:00 | Participants prepare and test their connectivity, policy, etc. |                                                   |
| 14:00 - 15:00 | Official competition: 1 Round of Off-Road Mission              | Official competition: 2 Rounds of Remote Missions |
| 15:00 - 16:00 | Official competition: 1 Round of Indoor Mission                |                                                   |
| 16:00 - 17:00 | Logistics Wrap Up                                              | Awards & Wrap Up                                  |

## Navigation Models Deployment

Competing teams shall host their own models in their own compute facilities while remotely accessing the assigned robots via a standard Remote Access SDK.

FrodoBots will also open-source a significant dataset of human tele-operated drives collected from 30+ cities (>7k hours), which the teams can opt to adopt as part of their models training pipeline.

There will be no limit on the size of the model as we hope to encourage teams to explore using large-scale foundational models. We may, however, further segment the competition into 2 separate categories based on compute size. This will be confirmed after taking in concrete feedback from participating teams.

## Models Testing Phase

In the months leading up to the actual competition, participants can test out their models on at least 2 open-world sites with pre-defined navigation missions ("seen environments"). We will choose at least 5 teams to participate in the 2-day competition during ICRA. During the actual competition, participants will attempt to complete navigation missions in unseen environments across at least 4 sites.



## Human Performance Benchmark

During the competition, human drivers (coming from the participating team members) will also attempt to complete the same missions alongside robots controlled by other participants' models. The human drivers are subject to the same conditions of “seen” vs “unseen” environments. This will help to form the human performance reference.

## Potential Target Participants

Google Deepmind, Meta AI, UC Berkeley, Georgia Tech, ETH, George Mason University, NUS, UT Austin, UMD, SNU, NYU, etc.

In terms of community reachout, we will create a competition website as well as various explanation/promotional videos. We will then actively outreach to the research community via existing email lists as well as social media platforms such as X/Twitter. The open-sourced driving dataset (>7k hours) will be hosted on Huggingface and we'll try working with them to amplify our message through their platform given their mindshare with the research community.

## Logistical Requests

We'd like to request an exhibition/teleoperation space of  $\sim 100\text{m}^2$  in the conference center for “Live Stream/Teleoperation” activities (described further below).

Finally, we'd like to explore working directly with any other local institutions who can provide their campus space as potential test sites.

**Live stream/Teleoperation:** Every ongoing drive (onsite or remotely, controlled by AI models or human drivers) will all be live-streamed in real-time.

As a rough estimate, if we have 10 participating teams and 8 robot locations/cities, we'd require at least 10 TV/projector displays just for live-streams, while the teams take turns to take on different missions.

We'd also like to setup live demos stations (here's what we did at another conference recently: <https://twitter.com/frodoBots/status/1750780293307646266> ) where ICRA attendees can try out driving these robots and completing some missions on the spot. This will further add on the requirement for TV/projectors.

Finally, we'd like to request for multiple dedicated internet ethernet lines with high bandwidth. This is critical as the live-streams and teleoperation controls will require stable high bandwidth to provide a great viewing/driving experience.

## Organizers

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Jie Tan (Google DeepMind)  
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Dhruv Shah (Princeton University)  
Xuesu Xiao (GMU)

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